

Networking System of a Hospital

1. Introduction

A hospital requires a reliable, secure, and scalable networking system to support day-to-day operations such as patient registration, doctor consultations, laboratory activities, pharmacy management, and centralized data storage. Efficient communication between departments is essential to deliver quality healthcare, ensure quick access to medical records, and maintain smooth workflow across the entire organization.

This project presents the design and implementation of a **Hospital Networking System** using **Cisco Packet Tracer**. The network includes multiple departments such as Reception, Doctor Room, Server Room, Laboratory, and Pharmacy. Each department is connected to a central multilayer switch, which connects to the main router and then to the cloud for external communication. The design ensures security, scalability, and high performance.

2. Objective

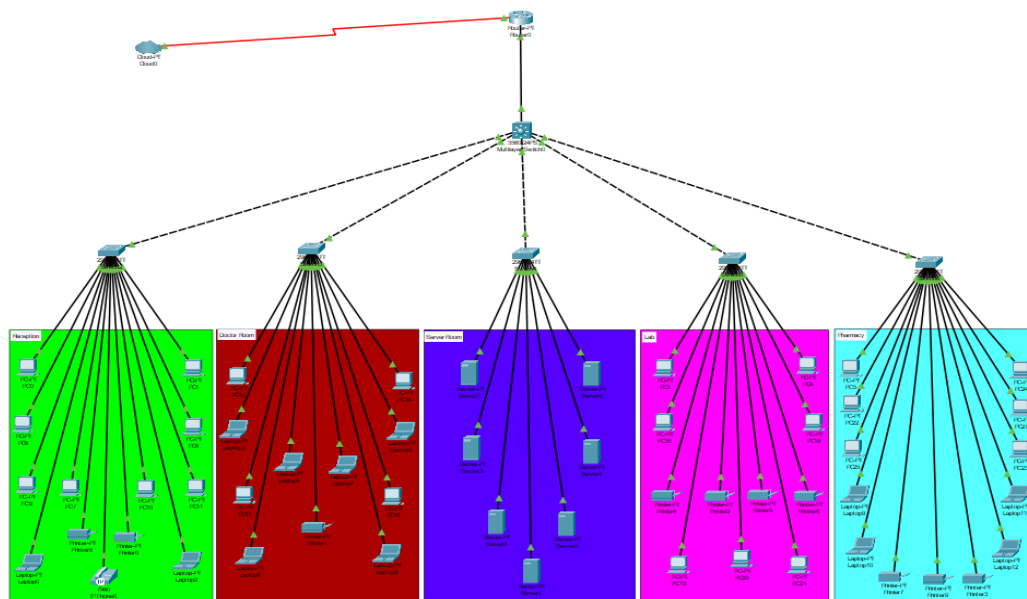
The main objectives of this project are:

1. To design and simulate a complete hospital network using Cisco Packet Tracer.
2. To provide reliable inter-department communication between Reception, Doctor Room, Lab, Pharmacy, and Server Room.
3. To ensure secure access to hospital data through centralized servers.
4. To implement a scalable network design that allows easy expansion of devices and departments.
5. To support essential hospital services such as patient records management, laboratory results, doctor prescriptions, and pharmacy operations.
6. To demonstrate proper use of networking devices (routers, switches, servers, PCs, and wireless devices).

3. Overview of the System

The hospital networking system is designed to connect all major departments—Reception, Doctor Room, Server Room, Laboratory, and Pharmacy—through a centralized and secure network infrastructure. At the core of the system, a multilayer switch serves as the main distribution point, linking each department's access switch while supporting VLAN segmentation and inter-VLAN routing to ensure secure and efficient data flow. The multilayer switch connects to a central router, which provides access to external networks through a cloud, enabling communication with online services or other healthcare facilities. Each department contains multiple PCs, laptops, printers, and IP phones connected to its local switch, allowing staff to perform tasks such as patient registration, record management, lab report handling, medical data storage, and pharmacy operations. The Server Room hosts essential servers, including application, database, and file servers, ensuring centralized storage and access to

electronic medical records. Overall, this system ensures smooth communication, fast data access, secure information sharing, and reliable operation across the entire hospital.



4. Requirements & Instruments

Hardware Requirements:

Device Type	Device Name	Quantity	Description
Router	Router-PT	1	Connects the hospital network to the Internet (via Cloud).
Cloud	Cloud-PT	1	Simulates external network/Internet connectivity.
Multilayer Switch	3560-24PS	1	Core switch for VLAN management and inter-department communication.
Switches	Switch-PT	5	Used in Reception, Doctor Room, Server Room, Lab, and Pharmacy.
PCs	PC-PT	37	Used by staff in all departments for data entry, record management, and communication.
Printers	Printer-PT	10	Departmental printers for report printing and prescriptions.
Servers	Server-PT	7	Central server for EMR, File Storage, DHCP, DNS, and Hospital Management System.

Software Requirements:

Category	Software / Tools	Purpose
Simulation Tool	Cisco Packet Tracer	Used to design, build, and test the hospital network.
Operating System	Windows / Linux	Used for documentation and running network utilities.
Network Protocols	TCP/IP, DHCP, DNS, VLAN, HTTP, HTTPS	Enables communication, addressing, server management, and web services.
Cabling	Copper Straight-through, Copper Cross-over	Used to connect PCs → Switches, Switches → Router, and inter-switch links.

Network Topology Summary:

Parameter	Details
Topology Type	Hierarchical Star Topology
Main Connection Flow	Cloud → Router → Multilayer Switch → Departmental Switches → PCs & Printers
Departments Connected	Reception, Doctor Room, Server Room, Laboratory, Pharmacy
Core Services Provided	Internet Access, File Sharing, Central Server Access, VLAN Communication, Data Management, Printing Services

5. Conclusion

The designed Hospital Networking System provides a complete, secure, and efficient communication structure for all hospital departments. It connects the Reception, Doctor Room, Laboratory, Pharmacy, and Server Room using a multilayer switch and a main router linked to a cloud. This architecture supports data management, patient record handling, lab reporting, medicine management, and essential hospital operations.

The system is scalable, allowing future expansion, and ensures high performance and network security. Using Cisco Packet Tracer helps visualize the architecture and understand the real-world implementation of such a healthcare network.